Solution architecture

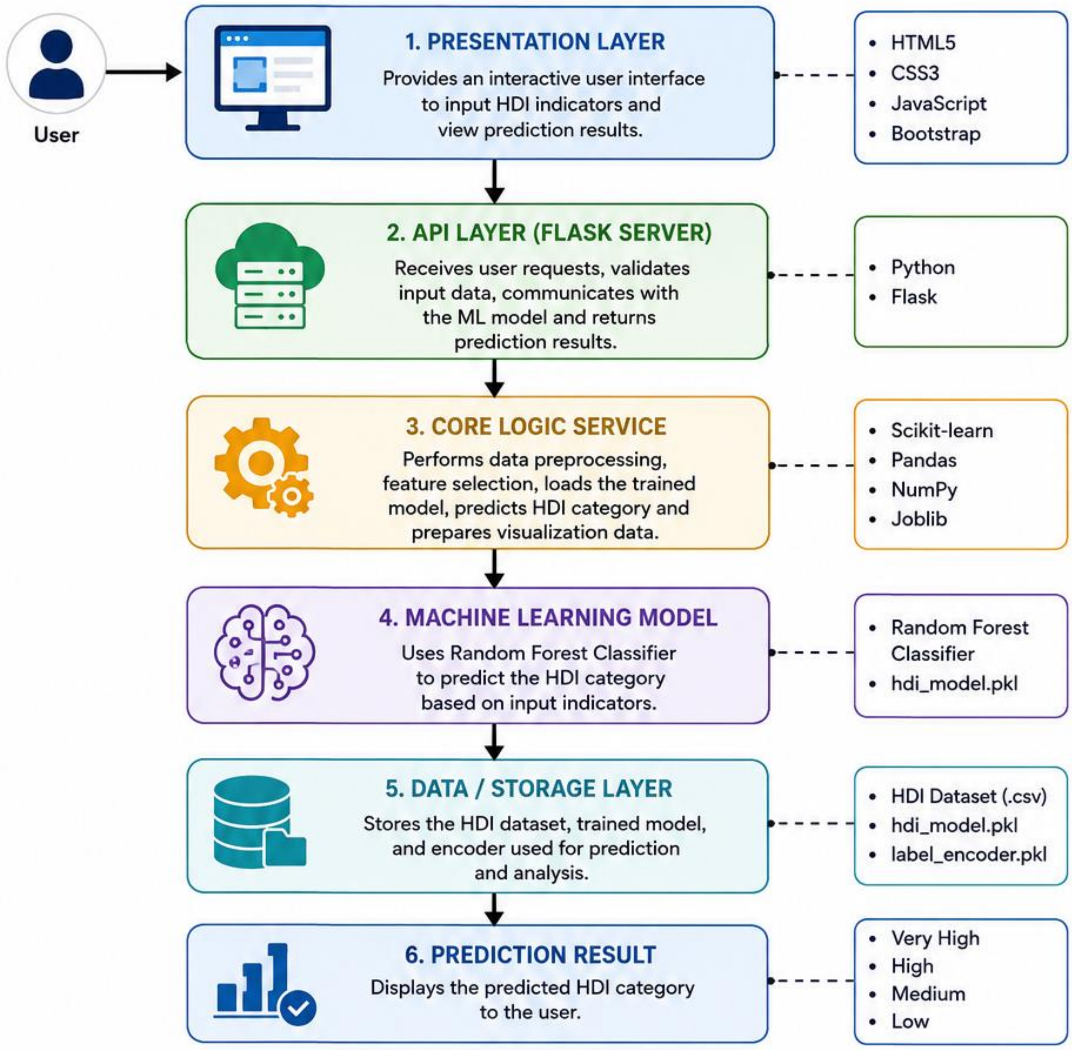
Date	05 JULY 2026
Team ID	XXXXX
Project Name	**A Comprehensive Measure of Well-Being**
Maximum marks	5 Marks

Solution architecture diagram:

SOLUTION ARCHITECTURE

Date: 05 JULY 2026	Team ID: XXXXX	Project Name: A Comprehensive Measure of Well-Being

SOLUTION ARCHITECTURE DIAGRAM



COMPONENT DESCRIPTION TABLE

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides an interactive user interface where users enter HDI indicators, view dashboards, graphs, and prediction results.	HTML5, CSS3, JavaScript, Bootstrap
API Layer (Flask Server)	Receives user requests, validates input data, communicates with the machine learning model, and returns prediction results.	Python, Flask
Core Logic Service	Performs data preprocessing, loads the trained machine learning model, predicts the HDI category, and prepares visualization data.	Scikit-learn, Pandas, NumPy, Joblib
Machine Learning Model	Predicts the Human Development Index category using the trained Random Forest Classification model.	Random Forest Classifier, Scikit-learn
Data / Storage Layer	Stores the HDI dataset and trained machine learning model used for prediction and analysis.	CSV Dataset, Pandas DataFrames, Pickle / Joblib
Prediction Output	Displays the predicted HDI category (Very High / High / Medium / Low) on the web application.	Flask, HTML, CSS